Tutorial 13

Exercise 1: Subset sum

Give a dynamic programming algorithm for Subset Sum. To this end, specify the recurrence relation (including initial values).

Exercise 2: Matrix multiplication

Multiplying an $n \times m$-matrix with an $m \times o$-matrix takes nmo multiplications of numbers using the naive matrix multiplication algorithm. Furthermore, matrix multiplication is associative, i.e., $A(BC) = (AB)C$ for matrices A, B, C of the right sizes. While the results of the different ways of multiplying the matrices are the same, the number of multiplications is not necessarily the same. For example, if A is a 10×30-matrix, B is a 30×5-matrix, and C is a 5×60-matrix, then

- computing $(AB)C$ takes 4500 multiplications while
- computing $A(BC)$ takes 27.000 multiplications.

The matrix multiplication problem is, given a sequence of n matrices $A_1, A_2, \ldots, A_n$ that can be multiplied as $A_1 A_2 \cdots A_n$, to determine the minimal number of number multiplications necessary to compute the product (but not to compute the product).

Give a dynamic programming algorithm for the matrix multiplication problem. To this end, specify the recurrence relation (including initial values).

Exercise 3: Challenge

Solve Day 16 of the Advent of Code 2022: `https://adventofcode.com/2022/day/16`. Use the input given on Moodle.